INCH-POUND
MIL-PRF-1/1341A
18 December 2008
SUPERSEDING
MIL-E-1/1341 (SigC)
1 April 1960

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY
ELECTROSTATIC DEFLECTION AND FOCUS
TYPE 3BGP1

Inactive for new design
after 5 December 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

ABSOLUTE RATING:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Light Output	Rg	Zd	ALt
Unit:	V	Vdc	V	Vdc	Vdc	ft. L	MegΩ	MegΩ	ft
Maximum	1.5 ± 10%	0	550	1100	2750	---	1.5	1.0	40,000
Minimum:	----	-200	----	----	1000 (Note 1)	20 (Note 2)	----	----	----
TEST CONDITIONS:	1.5	Adjust	----	focus	2000	----	----	---	------

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TABLE I. Testing and inspection.

Requirement or Test	Notes	Conditions	Acceptance level (See note 10)	Symbol	Limits		Unit
					Minimum	Maximum	
General							
Preproduction Sample Approval	3						
Performance							
Holding period							
Cathode-Ray tubes (Preheating)							
Dimensions (for cathode Ray tubes) (see figure 1)							
Container drop (Cathode Ray tubes)							
Preproduction Sample Approval Inspection	3						
Base connections							
Pressure		45 lb/sq. in					
Cathode-Ray vibration				Width	---	1.0	mm
Direct Interelectrode Capcitance		g1 to all D1 to D2 D3 to D4 D1 to all D2 to all D3 to all D4 to all	---- ---- ----- ---- ---- ---- ----	C C C C C C C	--- --- --- --- --- --- ---	6 4 4 7.5 7.5 6.0 6.0	µf µf µf µf µf µf µf
Cathode Illumination							
Acceptance Inspection Part 1							
Electride Currents Anode No. 1 Current		EcL=0		Ib1	-15	10	µA dc
Cathode current		Light = 20 ft.L		Ik	---	1000	µA dc
Voltage breakdown Voltage breakdown (electrostatic types)		Ef = 1.65 V		---	---	--- --	---
Gas “ Cross”		Light = 20 ft.L	---	---		----	---
Blemishes		---		---	---		
light outout	2		(See note 4)		Light 20	---	ft. L
Modulation		Light = 20 ft.L		∆Ec1		38	Vdc
Spot position (electrostatic deflection)						15.0	mm
Spot displacement (Leackage)				Displ	--	7.0	mm

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TABLE I. Testing and inspection. Continued

Requirement or Test	Notes	Conditions	Acceptance level (See note 10)	Symbol	Limits		Unit
					Minimum	Maximum	
Acceptance Inspection Part 1 continued							
Grid Cutoff voltage				Ec1	-38	-67.8	Vdc
Focusing Voltage at Cutoff		Ecl = Cutoff		Eb1	---	700	Vdc
Focusing Voltage, Zero-bias		Ecl = 0		Eb1	400	----	Vdc
Grid No. 1 Leakage		---		---	---	--	--
Anode No. 2 Leackage		---		----	---	---	---
Acceptance inspection Part 2 (design)							
Heater or filament current				If	126	154	mA
Alinement, Base (electrostatic types)		+3D4; Pin No. 5	6.5 6.5	----	-----	----	----
Alinement, side terminal		+1D2	6.5				
Alinement, side terminal and base		Pin No. 3	6.5				
Alinement, Neck and bulb			6.5	Diam	--	15	in
Alinement, Neck and base			6.5	---	---		----
Angle between traces			6.5	---	89	91	Deg
Angle, bulb and traces		+1D2; Bulb Wall	6.5		---	1.5	Deg
Stray Emission		Eb2 = 2750 Vdc	6.5	---	---	----	----
Line width “A’		Light = 20 ft. L	6.5	Width	---	0.65	mm
Line Width “D” Useful scan	5	Light = 20 ft. L	6.5	-----	-----	-----	mm
Deflection factor		1D2	6.5	IF	68	92	Vdc/in
Defection Factor		3D4	6.5	DF	28	38	Vdc/in
Deflection Factor Uniformity			6.5				

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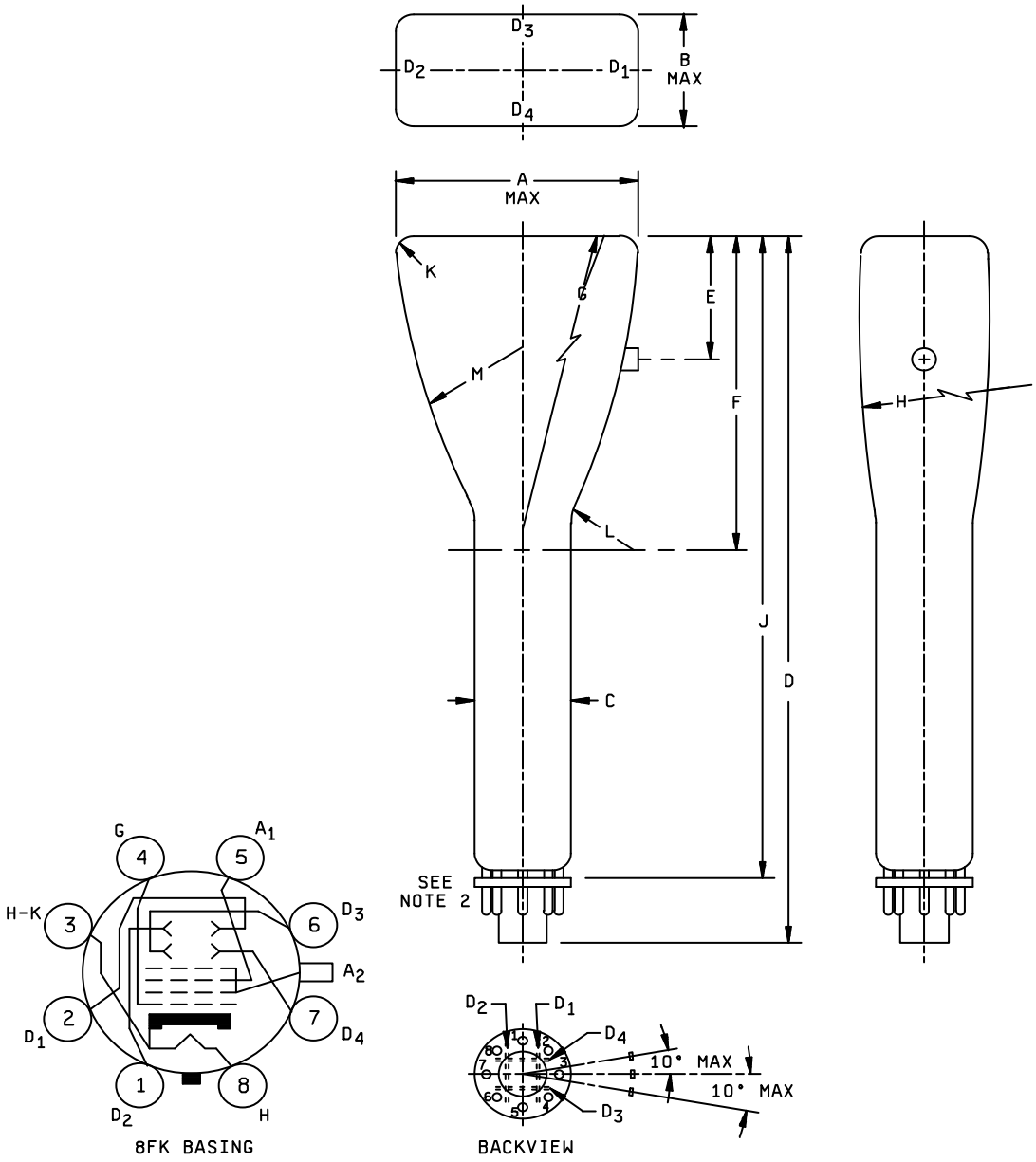
TABLE I. Testing and inspection. Continued

Requirement or Test	Notes	Conditions	Acceptance level (See note 10)	Symbol	Limits		Unit
					Min	Max	
Acceptance inspection Part 3 (Life)			---				
Life test		Group D; Light =20 ft. L; Eb2= 2750 Vdc	---	t	5 W	---	hrs
Life test End Point Line Width “A” Line Width “B” Modulation		 Light = 16ft.L Light = 16 ft. L Light = 16ft. L	----	 Width Width ΔEc	 --- --- ---	 0.65 0.65 38	 mm mm Vdc
Preparation for delivery	6		----	----	---	---	----

NOTES:

1. This value is recommended only for low velocity deflecting and low ambient light levels.
2. All light-dependent tests to be measured with a 1-1/8 inch vertical end 1-9/16 inch horizontal, 35 to 105 line raster. Photocell is to have one-inch active aperture and can be calibrated in foot-lamberts, similar to Photovolt Corporation Type 200-A or equivalent.
3. All tests applicable herein shall be performed during Preproduction Sample Approval Inspection; however, these 5 tests normally are performed during Preproduction Sample Approval Inspection only.
4. The acceptance level for the combined defectives attributes in acceptance inspection, part 1 (production), excluding mechanical and blemishes, shall be one percent.
5. The minimum useful scan I the major dimension D1D2, shall be 2-3/4 inches. The minimum useful scan in the minor direction, D3D4, shall be 1-1/8 inches.
6. Tubes shall be prepared for domestic or overseas shipment, as specified in the contract or order, in accordance with Specification MIL-DTL-75.
7. Identification marking of product shall be in accordance with MIL-PRF-1, except for the following
    a. Manufacturer’s qualification code-letter shall not be used in the product identification marking.
    b. If any specification requirements waiver has been granted, the product-identification marking shall consist of the type number only
8. The Preproduction Sample Approval requirements in Electronics Command Drawing ECOM- DWG SC-A-46600 hereby replace any Qualification requirements referable to the product covered herein (See Note 9 below).
9. Information pertaining to Preproduction Sample Approval testing may be obtained from Defense Supply Center Columbus, Code DSCC-VQE P.O. Box 3990, Columbus, OH 43216-5000. See the contracting officer for copies of Electronics Command Drawing SC-A-46600.
10. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

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Note

1: The minimum useful screen area shall be a rectangle 2.75 inches by 1.125 inches centered on the geometric center of the face and with the sides of the rectangle parallel to the axes of deflection.

2. The base be JEDEC B8-181.

FIGURE 1. Outline drawing of electron tube

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Ltr	Dimensions are in inches with metric equivalents (mm) in parentheses.	
	Mimimum	Maximum
Quality conformance inspection, part 2		
A (see note 1)	3.047 (77.40)	2.984 (75.79)
B (see note 1)	1.516 (38.51)	1.469 (37.31)
C	1.156 (39.62)	1.094 (27.78)
D	9.375 (238.12)	9.125 (231.77)
E	1.625 (37.31)	1.375 (41.27)
F	4.15 (104.77)	
G	60R	
H	30R	
J	8.3125 (211.13)	
K	.0934 (2.372)	
L	1R	
M	10 R	
N	.937 (23.7998)	

NOTES:
1. The minimum useful screen area shall be a rectangle 2.75 inches by 1.125 inches centered on the geometric center of the face and with the sides of the rectangle parallel to the axes of deflection
2. The base be JEDEC B8-181.

FIGURE 1. Outline drawing and dimensions continue

Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-DTL-75
ECOM- DWG SC-A-46600

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR
DLA – CC

Preparing activity:
DLA- CC

Project number (5960-2008-073)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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